CELLSCOPE DRIVE-TEST & RF COVERAGE REPORT

Automated Native Android Cellular Field Assessment • Lead: Sarkar Tuhassinul Arnob

1. Session Executive Summary

Session Name:	Survey_20261018_131911	**Start Timestamp:**	2026-10-18 13:19:11 (BST)
Duration:	00:00:02	**Mobility Mode:**	DYNAMIC DRIVE
Route Distance:	21.82 km	**Max Speed:**	63.7 km/h
Sample Count:	2023 measurements	**Mean RSRP:**	-88.6 dBm
GPS Availability:	100.0% (2023 valid)	**Radio Availability:**	100.0%
Coverage Adequacy:	87.9%	**Serving Cells Observed:**	44 distinct PCIs

2. RF Signal Quality Distribution (3GPP TS 36.133)



Excellent (≥ -80 dBm) **13.1% (266)**

Good (-90 to -81 dBm) **36.8% (744)**

Fair (-100 to -91 dBm) **33.8% (684)**

Poor (< -100 dBm) **16.3% (329)**

3. Handover & Serving-Cell Dwell Analytics

Total Cell Transitions: 86 | Confirmed Handovers: 33 | Reselections: 4

Transition Density: 3.94 transitions/km | Signal Loss Reselections: 0

PCI	Samples	Dwell Time	Session Share	Mean RSRP
PCI 347	161	2m 47s	7.9%	-95.1 dBm
PCI 41	137	2m 22s	6.7%	-94.5 dBm
PCI 6	128	2m 13s	6.3%	-107.9 dBm
PCI 345	112	1m 56s	5.5%	-96.4 dBm
PCI 426	106	1m 51s	5.3%	-89.0 dBm
PCI 48	103	1m 47s	5.1%	-94.1 dBm
PCI 61	94	1m 40s	4.8%	-68.4 dBm
PCI 43	92	1m 36s	4.6%	-84.3 dBm
PCI 35	81	1m 24s	4.0%	-93.5 dBm
PCI 67	79	1m 22s	3.9%	-89.9 dBm
PCI 329	71	1m 13s	3.5%	-85.0 dBm
PCI 389	70	1m 12s	3.4%	-88.1 dBm
PCI 19	68	1m 10s	3.3%	-85.0 dBm
PCI 20	64	1m 07s	3.2%	-83.8 dBm
PCI 26	65	1m 07s	3.2%	-82.8 dBm
PCI 39	61	1m 03s	3.0%	-94.0 dBm
PCI 485	60	1m 02s	3.0%	-76.1 dBm
PCI 367	46	0m 48s	2.3%	-89.2 dBm
PCI 13	44	0m 45s	2.2%	-85.7 dBm
PCI 217	41	0m 42s	2.0%	-94.0 dBm
PCI 265	40	0m 42s	2.0%	-82.5 dBm
PCI 25	40	0m 41s	2.0%	-97.5 dBm
PCI 40	29	0m 30s	1.4%	-90.0 dBm
PCI 66	28	0m 28s	1.4%	-90.0 dBm
PCI 264	21	0m 21s	1.0%	-80.0 dBm
PCI 241	20	0m 20s	1.0%	-89.0 dBm
PCI 250	20	0m 20s	1.0%	-44.0 dBm
PCI 153	19	0m 19s	0.9%	-95.0 dBm
PCI 5	17	0m 17s	0.8%	-70.5 dBm
PCI 62	17	0m 17s	0.8%	-80.8 dBm
PCI 266	14	0m 15s	0.7%	-89.0 dBm
PCI 453	10	0m 10s	0.5%	-83.0 dBm
PCI 143	9	0m 09s	0.4%	-105.0 dBm
PCI 203	8	0m 08s	0.4%	-44.0 dBm
PCI 29	8	0m 08s	0.4%	-107.0 dBm

PCI	Samples	Dwell Time	Session Share	Mean RSRP
PCI 178	7	0m 07s	0.3%	-62.6 dBm
PCI 18	6	0m 06s	0.3%	-44.0 dBm
PCI 53	6	0m 06s	0.3%	-44.0 dBm
PCI 319	6	0m 06s	0.3%	-103.5 dBm
PCI 173	5	0m 05s	0.2%	-44.0 dBm
PCI 366	4	0m 04s	0.2%	-44.0 dBm
PCI 447	3	0m 03s	0.1%	-44.0 dBm
PCI 491	2	0m 02s	0.1%	-108.0 dBm
PCI 328	1	0m 01s	0.0%	-86.0 dBm

4. Identified Weak-Coverage Zones & Remediation Feasibility

Zone #1: Length ~261m | Mean RSRP: -108.8 dBm | Min: -110 dBm
Diagnosis: Moderate Shadowing & Cell Edge: Signal fading near serving boundary (PCI 29, PCI 67).
Recommended Action: [ANTENNA_OPTIMIZATION] Sector Power Boost / Tilt Refinement (Feasibility: 75%)
Rationale: Adjusting RS (Reference Signal) power allocation or mechanical tilt may extend effective coverage radius into this edge zone.

Zone #2: Length ~64m | Mean RSRP: -106.9 dBm | Min: -107 dBm
Diagnosis: Moderate Shadowing & Cell Edge: Signal fading near serving boundary (PCI 6, PCI 319).
Recommended Action: [ANTENNA_OPTIMIZATION] Sector Power Boost / Tilt Refinement (Feasibility: 75%)
Rationale: Adjusting RS (Reference Signal) power allocation or mechanical tilt may extend effective coverage radius into this edge zone.

Zone #3: Length ~81m | Mean RSRP: -109.0 dBm | Min: -109 dBm
Diagnosis: Moderate Shadowing & Cell Edge: Signal fading near serving boundary (PCI 178, PCI 20).
Recommended Action: [ANTENNA_OPTIMIZATION] Sector Power Boost / Tilt Refinement (Feasibility: 75%)
Rationale: Adjusting RS (Reference Signal) power allocation or mechanical tilt may extend effective coverage radius into this edge zone.

Zone #4: Length ~208m | Mean RSRP: -107.0 dBm | Min: -108 dBm
Diagnosis: Moderate Shadowing & Cell Edge: Signal fading near serving boundary (PCI 26, PCI 6).
Recommended Action: [ANTENNA_OPTIMIZATION] Sector Power Boost / Tilt Refinement (Feasibility: 75%)
Rationale: Adjusting RS (Reference Signal) power allocation or mechanical tilt may extend effective coverage radius into this edge zone.

Zone #5: Length ~464m | Mean RSRP: -105.9 dBm | Min: -108 dBm
Diagnosis: Moderate Shadowing & Cell Edge: Signal fading near serving boundary (PCI 491, PCI 6, PCI 345).
Recommended Action: [ANTENNA_OPTIMIZATION] Sector Power Boost / Tilt Refinement (Feasibility: 75%)
Rationale: Adjusting RS (Reference Signal) power allocation or mechanical tilt may extend effective coverage radius into this edge zone.

Zone #6: Length ~1279m | Mean RSRP: -107.7 dBm | Min: -108 dBm
Diagnosis: Moderate Shadowing & Cell Edge: Signal fading near serving boundary (PCI 6, PCI 48).
Recommended Action: [ANTENNA_OPTIMIZATION] Sector Power Boost / Tilt Refinement (Feasibility: 75%)
Rationale: Adjusting RS (Reference Signal) power allocation or mechanical tilt may extend effective coverage radius into this edge zone.

Zone #7: Length ~73m | Mean RSRP: -106.0 dBm | Min: -106 dBm
Diagnosis: Moderate Shadowing & Cell Edge: Signal fading near serving boundary (PCI 40).
Recommended Action: [ANTENNA_OPTIMIZATION] Sector Power Boost / Tilt Refinement (Feasibility: 75%)
Rationale: Adjusting RS (Reference Signal) power allocation or mechanical tilt may extend effective coverage radius into this edge zone.

Zone #8: Length ~206m | Mean RSRP: -108.0 dBm | Min: -108 dBm
Diagnosis: Moderate Shadowing & Cell Edge: Signal fading near serving boundary (PCI 35).
Recommended Action: [ANTENNA_OPTIMIZATION] Sector Power Boost / Tilt Refinement (Feasibility: 75%)
Rationale: Adjusting RS (Reference Signal) power allocation or mechanical tilt may extend effective coverage radius into this

edge zone.

Zone #9: Length ~204m | Mean RSRP: -106.7 dBm | Min: -108 dBm
Diagnosis: Moderate Shadowing & Cell Edge: Signal fading near serving boundary (PCI 6, PCI 143).
Recommended Action: [ANTENNA_OPTIMIZATION] Sector Power Boost / Tilt Refinement (Feasibility: 75%)
Rationale: Adjusting RS (Reference Signal) power allocation or mechanical tilt may extend effective coverage radius into this edge zone.

5. Explainable AI Network Anomaly Detection (Candidate Hypotheses)

Identified 61 candidate anomalies (42 Critical, 19 Warning, 0 Advisory). ML consensus: 100.0%.

- **[WARNING] Rapid Handover Ping-Pong (PCI 18 ⇄ 61)**
  Hypothesis: Candidate inadequate handover hysteresis / Time-To-Trigger (TTT) parameterization or overlapping coverage boundaries with rapid fading.
- **[WARNING] Rapid Handover Ping-Pong (PCI 61 ⇄ 18)**
  Hypothesis: Candidate inadequate handover hysteresis / Time-To-Trigger (TTT) parameterization or overlapping coverage boundaries with rapid fading.
- **[WARNING] Abrupt Signal Drop (-25 dB in 4s)**
  Hypothesis: Candidate sharp shadow fading from overhead flyover, tunnel entrance, reinforced concrete structure, or antenna feeder defect.
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  Hypothesis: Candidate sharp shadow fading from overhead flyover, tunnel entrance, reinforced concrete structure, or antenna feeder defect.

6. Co-Channel, Adjacent-Channel & PCI Reuse Analysis

Evaluated 515 co-channel allocations, 0 adjacent carrier allocations, and 187 CRS modulo-3 collisions. Correlation: Identified 234 potential co-channel/CRS subcarrier contention pair(s). SINR metric not exposed by device modem; verification requires field scanner or network trace.

- **[Potential PCI Reuse Concern] Ch 1399 | PCI 18 & 18 (Sep: 121m)**
  Identical PCI 18 reused on channel 1399 with spatial separation of 121m (<2000m). Potential uncoordinated PCI collision hazard.
- **[Potential PCI Reuse Concern] Ch 1399 | PCI 61 & 61 (Sep: 162m)**
  Identical PCI 61 reused on channel 1399 with spatial separation of 162m (<2000m). Potential uncoordinated PCI collision hazard.
- **[Potential PCI Reuse Concern] Ch 1399 | PCI 20 & 20 (Sep: 225m)**
  Identical PCI 20 reused on channel 1399 with spatial separation of 225m (<2000m). Potential uncoordinated PCI collision hazard.

Interference Limitation: Note: Radio interference analysis correlates upper-layer Android TelephonyManager cell observations (carrier frequency and nominal bandwidth). Standard modems do not expose raw ACLR, spectral leakage, or RF emissions masks. Classifications indicate allocation geometry and co-channel proximity hypotheses rather than direct baseband spectrum measurements.

7. Multi-Generation Network Coverage, Continuity & Service Hole Intelligence

5G NR: 0 (0.0%) | 4G LTE: 2023 (100.0%) | 3G: 0 (0.0%) | 2G: 0 (0.0%) | OOS: 0 (0.0%)

Continuity Events: 0 total shifts (0 downgrades, 0 service losses). Generation transitions are recorded with engineering evidence without assuming unconfirmed causes.

Service Holes: 3 events identified across 7 distinct outage categories (CELLULAR_DATA_DISCONNECTED: 3).

- **[CELLULAR_DATA_DISCONNECTED] Severity: HIGH at 13:21:54.745 (BST) (1s duration)**
  Radio in service but cellular data packet bearer reported disconnected for 1.2 seconds.
- **[CELLULAR_DATA_DISCONNECTED] Severity: HIGH at 13:29:30.019 (BST) (19s duration)**
  Radio in service but cellular data packet bearer reported disconnected for 19.9 seconds.

- **Voice Call Reliability & Drop Risk Score:** **LOW (Risk Score: 0/100)**
  Evaluation: 5 call events evaluated (0 suspected drops). Explanation: Adequate cellular coverage stability. Call-drop risk indicators remain low along the tested route.

Multi-Gen Measurement Note: Voice and data bearers are independently evaluated. Telemetry gaps and test timeouts are strictly distinguished from true radio outages.

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8. Methodology & Device Limitations Note:
Measurements captured via Android TelephonyManager and LocationManager. Telephony polling frequency, neighbor cell reporting, and SINR availability are device-, baseband- and carrier-dependent. Coverage surfaces and tower positions are mathematical inferences.